

CBSE 2027 Chemistry — Predicted Paper (Set 8 of 10)

Based on the highly repeated questions from previous years' board exams (PYQs), official CBSE directives, and the most critical NCERT concepts, here is Set 8 of the predicted board paper [1, 2]. All questions are entirely new and distinct from the previous sets, targeting heavily tested formulas, name reactions, and analytical reasoning [3, 4].

SECTION A: Multiple Choice & Assertion-Reason Questions (1 Mark Each x 16 = 16 Marks)

Q1. The value of the ebullioscopic constant (K_b) depends exclusively on: (a) the nature of the solute (b) the nature of the solvent (c) the temperature of the solution (d) the concentration of the solution * **Chapter:** Solutions | **Confidence:** Very High [5]

Q2. Which of the following halogens does NOT exhibit a positive oxidation state? (a) Fluorine (b) Chlorine (c) Bromine (d) Iodine * **Chapter:** p-Block Elements (or d- and f-Block general properties) | **Confidence:** High [6]

Q3. The rate law for a reaction is given by $\text{Rate} = k[A]^1[B]^2$. If the concentration of reactant B is doubled while keeping A constant, the rate of the reaction will: (a) double (b) remain the same (c) increase by four times (d) increase by eight times * **Chapter:** Chemical Kinetics | **Confidence:** Very High [7]

Q4. The transition metal with the lowest enthalpy of atomization in the 3d series is: (a) Scandium (Sc) (b) Chromium (Cr) (c) Manganese (Mn) (d) Zinc (Zn) * **Chapter:** d- and f-Block Elements | **Confidence:** High [8]

Q5. In the coordination complex $[Pt(en)_2Cl_2]^{2+}$, the coordination number and oxidation state of Platinum are respectively: (a) 6 and +4 (b) 4 and +2 (c) 6 and +2 (d) 4 and +4 * **Chapter:** Coordination Compounds | **Confidence:** Very High [9]

Q6. Alkyl halides react with metallic sodium in the presence of dry ether to form alkanes containing double the number of carbon atoms. This specific reaction is known as: (a) Fittig reaction (b) Wurtz-Fittig reaction (c) Wurtz reaction (d) Sandmeyer reaction * **Chapter:** Haloalkanes and Haloarenes | **Confidence:** Very High [10]

Q7. Which reagent is used to convert phenol into 2,4,6-tribromophenol, yielding a white precipitate? (a) Br_2 in CS_2 (b) Bromine water (Br_2/H_2O) (c) HBr and heat (d) Br_2 in anhydrous $AlCl_3$ * **Chapter:** Alcohols, Phenols and Ethers | **Confidence:** Very High [11]

Q8. The reaction of a primary amide with bromine and aqueous sodium hydroxide to yield an amine with one less carbon atom is called: (a) Gabriel Phthalimide Synthesis (b) Hoffmann Bromamide Degradation (c) Carbylamine Reaction (d) Sandmeyer Reaction * **Chapter:** Amines | **Confidence:** Very High [12, 13]

Q9. Which of the following is a condensation polymer? (a) Teflon (b) PVC (c) Polythene (d) Bakelite * **Chapter:** Polymers (Concept integrated in Organics) | **Confidence:** Medium [14]

Q10. The nitrogenous base present in RNA but absent in DNA is: (a) Adenine (b) Guanine (c) Thymine (d) Uracil * **Chapter:** Biomolecules | **Confidence:** Very High [15]

Q11. The Swarts reaction is specifically used for the synthesis of: (a) Alkyl chlorides (b) Alkyl bromides (c) Alkyl iodides (d) Alkyl fluorides * **Chapter:** Haloalkanes and Haloarenes | **Confidence:** Very High [16]

Q12. According to Valence Bond Theory (VBT), the hybridization of the central metal ion in $[Ni(CN)_4]^{2-}$ is: (a) sp^3 (b) dsp^2 (c) sp^3d^2 (d) d^2sp^3 * **Chapter:** Coordination Compounds | **Confidence:** High [17]

Directions for Q13 to Q16: Select the correct answer from the codes (a), (b), (c), and (d).

(a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is NOT the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true.

Q13. Assertion (A): The molecularity of a complex, multi-step reaction has no meaning. **Reason (R):** A complex reaction takes place in several elementary steps, and molecularity is only defined for individual elementary steps. * **Chapter:** Chemical Kinetics | **Confidence:** Very High [18]

Q14. Assertion (A): Benzylamine is more basic than aniline. **Reason (R):** In aniline, the lone pair of electrons on the nitrogen atom is delocalized over the benzene ring, making it less available for protonation. * **Chapter:** Amines | **Confidence:** Very High [19, 20]

Q15. Assertion (A): Ce^{4+} is an excellent analytical oxidizing agent. **Reason (R):** Cerium (IV) readily gains an electron to attain the highly stable +3 oxidation state, which is the most common oxidation state for lanthanoids. * **Chapter:** d- and f-Block Elements | **Confidence:** High [21]

Q16. Assertion (A): Phenol forms a yellow precipitate with 2,4-DNP reagent. **Reason (R):** Phenol is highly acidic compared to aliphatic alcohols. * **Chapter:** Alcohols, Phenols and Ethers / Aldehydes & Ketones | **Confidence:** High [22, 23]

SECTION B: Very Short Answer Questions (2 Marks Each x 5 = 10 Marks)

Q17. (i) Define the term *molality* (m). (ii) Why is molality preferred over molarity for expressing the concentration of a solution when dealing with temperature variations? * **Chapter:** Solutions | **Confidence:** Very High [24]

Q18. Differentiate between *homoleptic* and *heteroleptic* coordination complexes. Support your answer with one chemical example of each. * **Chapter:** Coordination Compounds | **Confidence:** High [25]

Q19. A radioactive substance follows first-order kinetics. If its half-life ($t_{1/2}$) is 12 hours, calculate the fraction of the initial amount of the substance remaining after exactly 36 hours. * **Chapter:** Chemical Kinetics | **Confidence:** Very High [26]

Q20. Arrange the following compounds in increasing order of their reactivity towards an S_N2 reaction mechanism. Give a brief reason for your order. 1-Bromopentane, 2-Bromopentane,

2-Bromo-2-methylbutane. * **Chapter:** Haloalkanes and Haloarenes | **Confidence:** Very High [27]

Q21. (i) Why do transition metals frequently form interstitial compounds? (ii) State one physical characteristic of interstitial compounds compared to their parent pure metals (e.g., in terms of hardness). * **Chapter:** d- and f-Block Elements | **Confidence:** High [28]

SECTION C: Short Answer Questions (3 Marks Each x 7 = 21 Marks)

Q22. Calculate the boiling point elevation (ΔT_b) and the boiling point of a solution prepared by adding 10 g of Calcium Chloride (CaCl_2) to 200 g of water. Assume that CaCl_2 undergoes complete dissociation in the solution. (Given: K_b for water = $0.512 \text{ K kg mol}^{-1}$, Molar mass of $\text{CaCl}_2 = 111 \text{ g mol}^{-1}$, Boiling point of pure water = 373.15 K) * **Chapter:** Solutions | **Confidence:** Very High [29]

Q23. The molar conductivity (Λ_m) of a 1.5 M solution of an electrolyte is found to be $138.9 \text{ S cm}^2 \text{ mol}^{-1}$. Calculate the conductivity (κ) of this solution. State the SI unit of conductivity. * **Chapter:** Electrochemistry | **Confidence:** Very High [30]

Q24. The rate constants of a chemical reaction at 200 K and 500 K are 0.02 s^{-1} and 0.20 s^{-1} , respectively. Calculate the activation energy (E_a) for this reaction. (Given: $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$, $\log 10 = 1$) * **Chapter:** Chemical Kinetics | **Confidence:** Very High [31]

Q25. Outline the necessary reagents and reaction steps to carry out the following organic conversions: (i) Propene to Propan-2-ol (ii) Phenol to Salicylaldehyde (iii) Benzyl chloride to Benzyl alcohol * **Chapter:** Alcohols, Phenols and Ethers | **Confidence:** Very High [32, 33]

Q26. Give simple, observable chemical tests to distinguish between the following pairs of compounds: (i) Acetophenone and Benzophenone (ii) Phenol and Benzoic acid (iii) Formic acid and Acetic acid * **Chapter:** Aldehydes, Ketones and Carboxylic Acids | **Confidence:** Very High [34, 35]

Q27. Answer the following biochemical questions: (i) What are reducing sugars? Give one example. (ii) What are the final products of the complete hydrolysis of maltose? (iii) Why must Vitamin C be supplied regularly in our diet? * **Chapter:** Biomolecules | **Confidence:** High [14, 36]

Q28. Account for the following observations regarding d-block elements: (i) Zinc (Zn), Cadmium (Cd), and Mercury (Hg) are comparatively soft metals. (ii) Transition metals exhibit a wide variety of oxidation states. (iii) The standard electrode potential (E°) value for the V^{3+}/V^{2+} couple is relatively low (negative). * **Chapter:** d- and f-Block Elements | **Confidence:** Very High [37, 38]

SECTION D: Case Study-Based Questions (4 Marks Each x 2 = 8 Marks)

Q29. Case Study 1: Commercial Batteries and Fuel Cells Batteries are practical applications of galvanic cells. They are broadly classified into primary batteries (non-rechargeable, e.g., Dry cell, Mercury cell) and secondary batteries (rechargeable, e.g., Lead storage battery, Ni-Cd cell). A lead storage battery consists of a lead anode and a grid of lead packed with lead dioxide (PbO_2) as the cathode. A 38% solution of sulfuric acid (H_2SO_4) serves as the electrolyte. When the battery is in use (discharging), both electrodes become coated with lead sulfate ($PbSO_4$), and the concentration of the acid decreases. (i) Which specific material acts as the anode in a fully charged lead storage battery? (1 Mark) (ii) What happens to the density of the sulfuric acid electrolyte during the discharging phase of the battery? (1 Mark) (iii) Write the separate half-cell reactions occurring at the anode and the cathode during the **discharging** of a lead storage battery. (2 Marks) * **Chapter:** Electrochemistry | **Confidence:** Very High [39, 40]

Q30. Case Study 2: Nucleophilic Addition in Carbonyl Compounds Aldehydes and ketones undergo nucleophilic addition reactions due to the polarization of the carbonyl bond ($>C=O$). A nucleophile attacks the electrophilic carbon atom perpendicular to the sp^2 hybridized plane, forming a tetrahedral alkoxide intermediate. Aldehydes are generally more reactive than ketones towards nucleophilic attack due to both steric and electronic reasons. Bulky alkyl groups in ketones hinder the approach of the nucleophile, and their electron-donating (+I) effect reduces the electrophilicity of the carbonyl carbon. (i) Name the specific type of chemical product formed when hydrogen cyanide (HCN) adds to the carbonyl group of an aldehyde. (1 Mark) (ii) Based on the passage, state the hybridization change that occurs at the carbonyl carbon during a nucleophilic addition reaction. (1 Mark) (iii) Arrange the following compounds in increasing order of their reactivity towards nucleophilic addition by HCN: Formaldehyde ($HCHO$), Acetaldehyde (CH_3CHO), Acetone (CH_3COCH_3), and Acetophenone ($C_6H_5COCH_3$). Briefly justify your order using the concepts of steric hindrance and the +I effect. (2 Marks) * **Chapter:** Aldehydes, Ketones and Carboxylic Acids | **Confidence:** Very High [41-43]

SECTION E: Long Answer Questions (5 Marks Each x 3 = 15 Marks)

Q31. (A) Organic Compounds: Amines and Conversions An organic compound (A) with the molecular formula C_3H_7NO on heating with aqueous sodium hydroxide and bromine gas undergoes a degradation reaction to form a foul-smelling primary amine (B) with the formula C_2H_7N . When compound (B) is treated with nitrous acid (HNO_2) at cold temperatures, it yields ethanol (C_2H_5OH) and liberates nitrogen gas with brisk effervescence. (i) Identify the structures and IUPAC names of compounds (A) and (B). (2 Marks) (ii) Name the specific chemical reaction involved in the conversion of (A) to (B). (1 Mark) (iii) Write the complete, balanced chemical equations for: a) The conversion of (A) to (B). b) The reaction of (B) with HNO_2 . (2 Marks) * **OR (B)** Provide the appropriate chemical equations and name the specific reactions for the following: (i) Carbylamine reaction (using aniline as the starting material). (ii) Gabriel Phthalimide Synthesis (showing the formation of ethanamine). (iii) Coupling reaction of benzenediazonium chloride with phenol. (iv) Acetylation of aniline using acetic anhydride.

(v) Conversion of aniline to chlorobenzene using $\text{Cu}_2\text{Cl}_2/\text{HCl}$. * **Chapter:** Amines | **Confidence:** Very High [44, 45]

Q32. (A) Coordination Compounds: Crystal Field & Valence Bond Theory (i) According to Valence Bond Theory (VBT), predict the type of hybridization, geometry, and magnetic property (paramagnetic or diamagnetic) of the hexafluoridocobaltate(III) complex ion, $[\text{CoF}_6]^{3-}$. (Given atomic number of Co = 27, and fluoride is a weak field ligand). (3 Marks) (ii) Why are tetrahedral complexes generally known to form high-spin complexes rather than low-spin complexes? Explain using the concept of crystal field splitting energy (Δ_t) and pairing energy (P). (2 Marks) * **OR (B)** (i) Write the fundamental postulates of Werner's Coordination Theory, differentiating clearly between primary and secondary valencies. (2 Marks) (ii) Write the systematic IUPAC names for the following coordination compounds: a) $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$ b) $[\text{Pt}(\text{NH}_3)_2\text{Cl}(\text{NO}_2)]$ (2 Marks) (iii) What is the denticity of the EDTA ligand? (1 Mark) * **Chapter:** Coordination Compounds | **Confidence:** Very High [17, 46]

Q33. (A) Electrochemistry: Faraday's Laws & Nernst Equation (i) State Michael Faraday's First Law of Electrolysis. (1 Mark) (ii) Calculate the mass of magnesium (Mg) metal produced at the cathode when a steady current of 5.0 Amperes is passed through molten magnesium chloride (MgCl_2) for exactly 1 hour. (Given: Atomic mass of Mg = 24 g mol^{-1} , $1\text{F} = 96500 \text{ C mol}^{-1}$). (2 Marks) (iii) Formulate the Nernst equation and calculate the Electromotive Force (EMF) of the following cell at 298 K:
 $\text{Al}_{(s)}|\text{Al}^{3+}(0.01 \text{ M})||\text{Ni}^{2+}(0.1 \text{ M})|\text{Ni}_{(s)}$ (Given standard reduction potentials: $E^\circ_{\text{Al}^{3+}/\text{Al}} = -1.66 \text{ V}$ and $E^\circ_{\text{Ni}^{2+}/\text{Ni}} = -0.25 \text{ V}$). (2 Marks) * **OR (B)** (i) Define limiting molar conductivity (Λ_m°). According to Kohlrausch's law, how can the limiting molar conductivity of a weak electrolyte like acetic acid (CH_3COOH) be determined using strong electrolytes? (2 Marks) (ii) Write the relationship between standard cell potential (E°_{cell}) and the standard Gibbs free energy change (ΔG°). (1 Mark) (iii) Calculate ΔG° and the equilibrium constant (K_c) for the following standard cell reaction at 298 K: $\text{Zn}_{(s)} + \text{Cu}_{(aq)}^{2+} \rightarrow \text{Zn}_{(aq)}^{2+} + \text{Cu}_{(s)}$ (Given: $E^\circ_{\text{cell}} = 1.10 \text{ V}$, $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$, $1\text{F} = 96500 \text{ C mol}^{-1}$). (2 Marks) * **Chapter:** Electrochemistry | **Confidence:** Very High [47-49]

CBSE Class 12 (2027) — AI-Predicted via PYQ/Blueprint Analysis. Probability-weighted guidance, not actual exam questions.